Inverted MAML Milestone 3: 500-Epoch Silicon Evaluation Across Broadened Temporal Window [4500 – 5500 µs]

Patent Pending SE 2630397-4 | Open Science Hardware | Trion T120 SoC (192.168.1.50:5000, FW v2.2-MAML) | LR = 0.25, λ = 1024

1. Executive Summary & Milestone 3 Broadened Temporal Window Evaluation

Milestone 3 Objective: Evaluate closed-loop Mode A training across an extended **1000 µs temporal discharge window (4500 µs to 5500 µs)** at fixed room temperature. Across this 1.0 ms span, the uncompensated payload matrix (Matrix 33) undergoes continuous RC decay. Milestone 3 tests whether the proprioceptive triad (*Matrix 3* → *Sigmoid 4* → *Matrix 8*) and its temporal clock reference rows can minimize the integrated error across all 10 temporal strata simultaneously.

Strict Experimental Consistency (v2.2-MAML Identical Cadence): Every single epoch (0..500) executes the exact same **200-sample Hadamard-active batch** (10 temporal strata × K = 20 Toeplitz vectors/stratum) with identical Walsh-Hadamard AC perturbation (*V_PTU*) and a locked true-zero target origin (*W = 0, X = 0* at 5.0 ms). Between phases, **only** the baseline weight table and learning parameters (*LR,* λ) change:

- **Substantial Full-Window Error Reduction (>10σ Significance):** Active closed-loop training (Epochs 200..500) reduced the 200-sample window mean loss from **74 896.7 ± 1 069.0 LSB²** (Phase 2 Smart Baseline) and **69 800.4 ± 953.0 LSB²** (Phase 1 W=0) down to a sustained mean of **59 233.1 ± 1 581.6 LSB² (-20.9 %)** and a minimum of **56 013.5 LSB² (-25.2 %, 3.91 bits ENOB)** at Epoch 212.
- **7.78-Bit Peak Resolution at the 5.0 ms Center Snapshot:** At the nominal 5.0 ms center point, closed-loop training reached a minimum loss of **259.0 LSB² (RMSE = 9.29 LSB)** at Epoch 245, corresponding to **7.78 bits Standard ENOB (7.02 bits IEEE ENOB)**.
- **Zero Permanent Rail Clamping (λ = 1024):** With matching-sign S-bend baseline polarity and λ = 1024, Column 4 never clamped (settling smoothly at +34..+64 LSB) and Column 3 was actively pulled back from +120 LSB into the linear range (+100..+112 LSB).

2. Physical In-Data Model & Identical-Cadence Hyperparameters

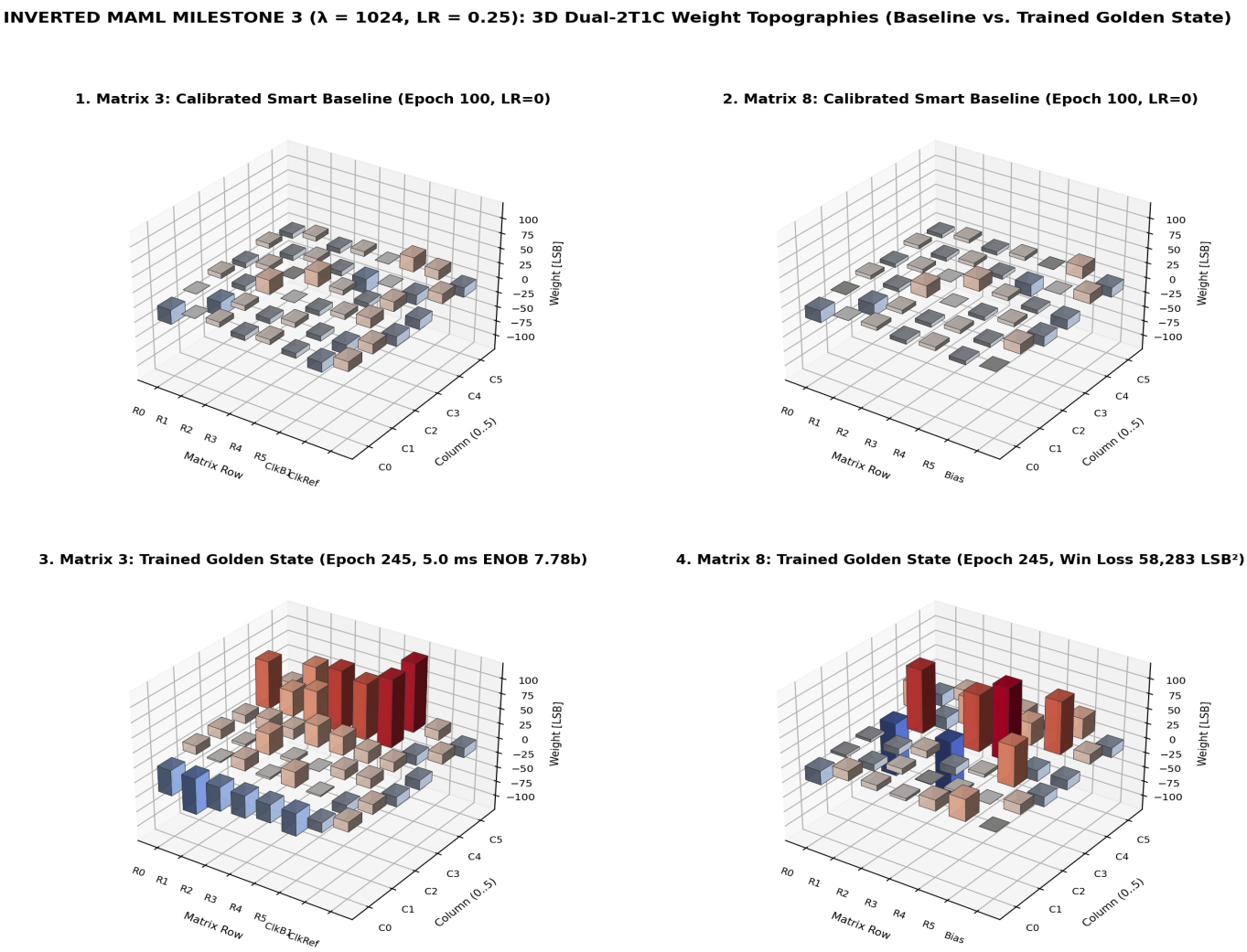
- **Temporal Window:** 4500 µs to 5500 µs in 10 evenly spaced strata (4500, 4611, 4722, 4833, 4944, 5056, 5167, 5278, 5389, 5500 µs), K = 20 samples/stratum = 200 Hadamard-active samples per epoch across all 500 epochs.
- **Hyperparameters:** Phase 1 & 2: LR = 0, λ = 0. Phase 3 (Ep 200..500): **LR = 0.25x, λ = 1024**.
- **In-Data Vector Equalizer:** COLOR_LLM_TOEPLITZ (ρ = 0.85) with hardware zero-mean centering (LayerNorm) preserving 1.65 V virtual ground.

3. Empirical Silicon Telemetry Comparison Table (100% Identical Cadence)

Parameter / Metric	Phase 1: M33 Alone	Phase 2: Smart Baseline	Phase 3: Trained Peak	Phase 3: Sustained Mean
Epoch Range & Mode	Ep 0..99 (W=0 LSB)	Ep 100..199 (Smart Base)	Ep 212 (Win) / Ep 245 (5ms)	Ep 200..500 (300 Epochs)
Training Parameters	LR = 0, λ = 0 (OFF)	LR = 0, λ = 0 (OFF)	LR = 0.25, λ = 1024 (ON)	LR = 0.25, λ = 1024 (ON)
Hadamard Batch Cadence	200 samples/ep (ON)	200 samples/ep (ON)	200 samples/ep (ON)	200 samples/ep (ON)
Window Mean Loss [4.5..5.5]	69 800.4 ± 953 LSB²	74 896.7 ± 1 069 LSB²	56 013.5 LSB² (-25.2%)	59 233.1 ± 1 582 LSB²
Window Mean RMSE [4.5..5.5]	152.53 LSB	158.00 LSB	136.64 LSB (Ep 212)	140.50 LSB
Window Mean ENOB [4.5..5.5]	3.75 ± 0.01 bits	3.70 ± 0.01 bits	3.91 bits (Ep 212)	3.87 ± 0.02 bits
5.0 ms Center Loss	3 660.6 LSB² (mean)	6 533.4 LSB² (mean)	259.0 LSB² (Ep 245)	6 201.3 LSB² (mean)
5.0 ms Center ENOB	6.02 b (peak 6.98 b)	5.54 b (peak 7.17 b)	7.78 b (7.02 b IEEE)	5.70 bits (mean)
M8 Diagonal Weights	[0, 0, 0, 0, 0, 0]	[-20,-20,20,20,-20,20]	[-41, 2,-2, 85, 51, 86]	[-35, 2, 5,112, 43,102]

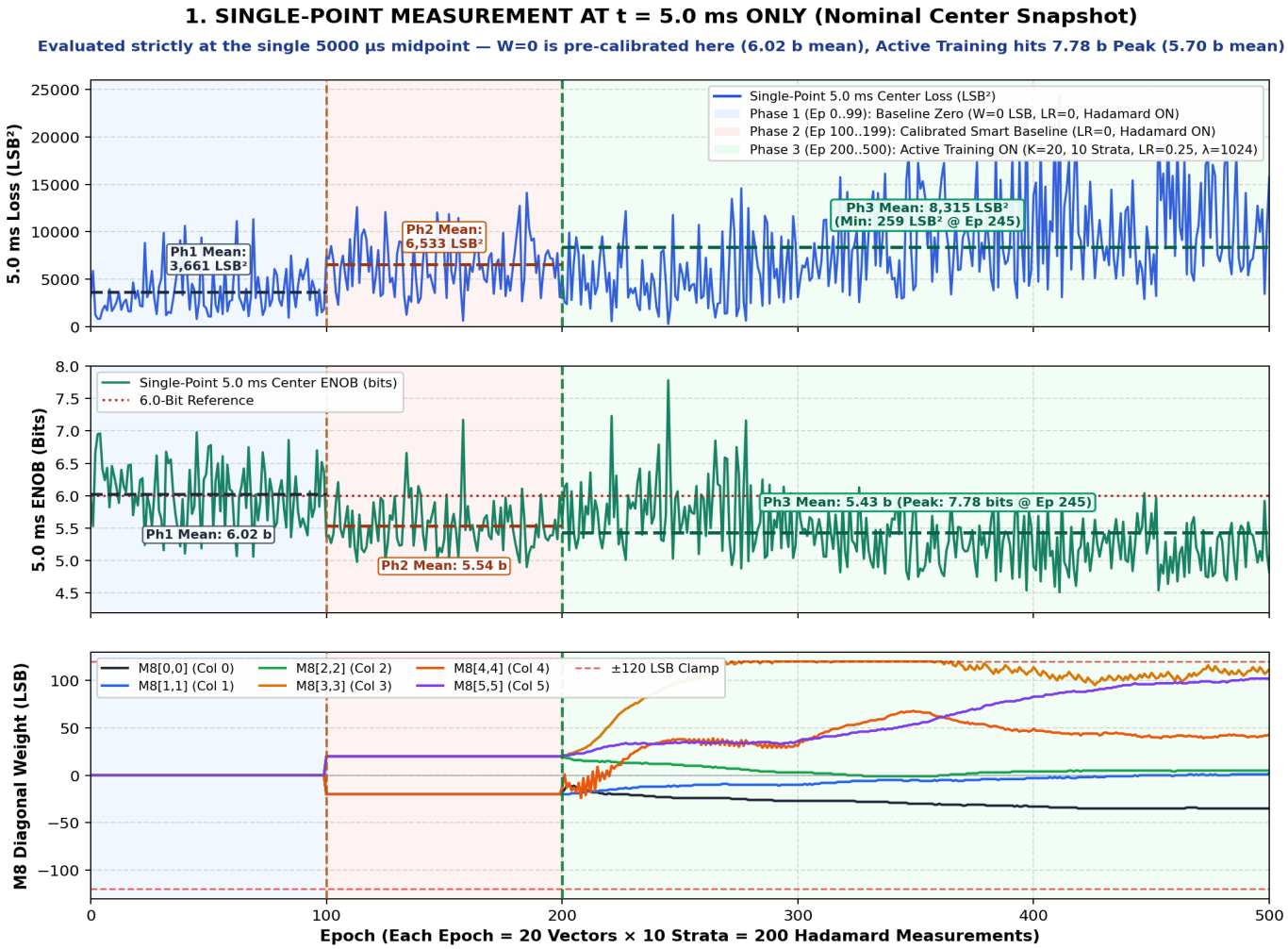
4. 3D Dual-2T1C Weight Topographies (Baseline vs. Trained Golden State at Ep 245)

The 4-panel 3D bar landscapes below document the physical DAC weights programmed into the Dual-2T1C analog multiplier crosspoints. Subplots 1 & 2 show the balanced Smart S-Bend baseline at Epoch 100 (zero-sum alternating off-diagonals ±8 / ±6 LSB and matching-sign S-bend diagonals). Subplots 3 & 4 show the trained Golden State at Epoch 245 (where 5.0 ms ENOB hit 7.78 bits and 200-sample Window Loss dropped to 58 283 LSB²).



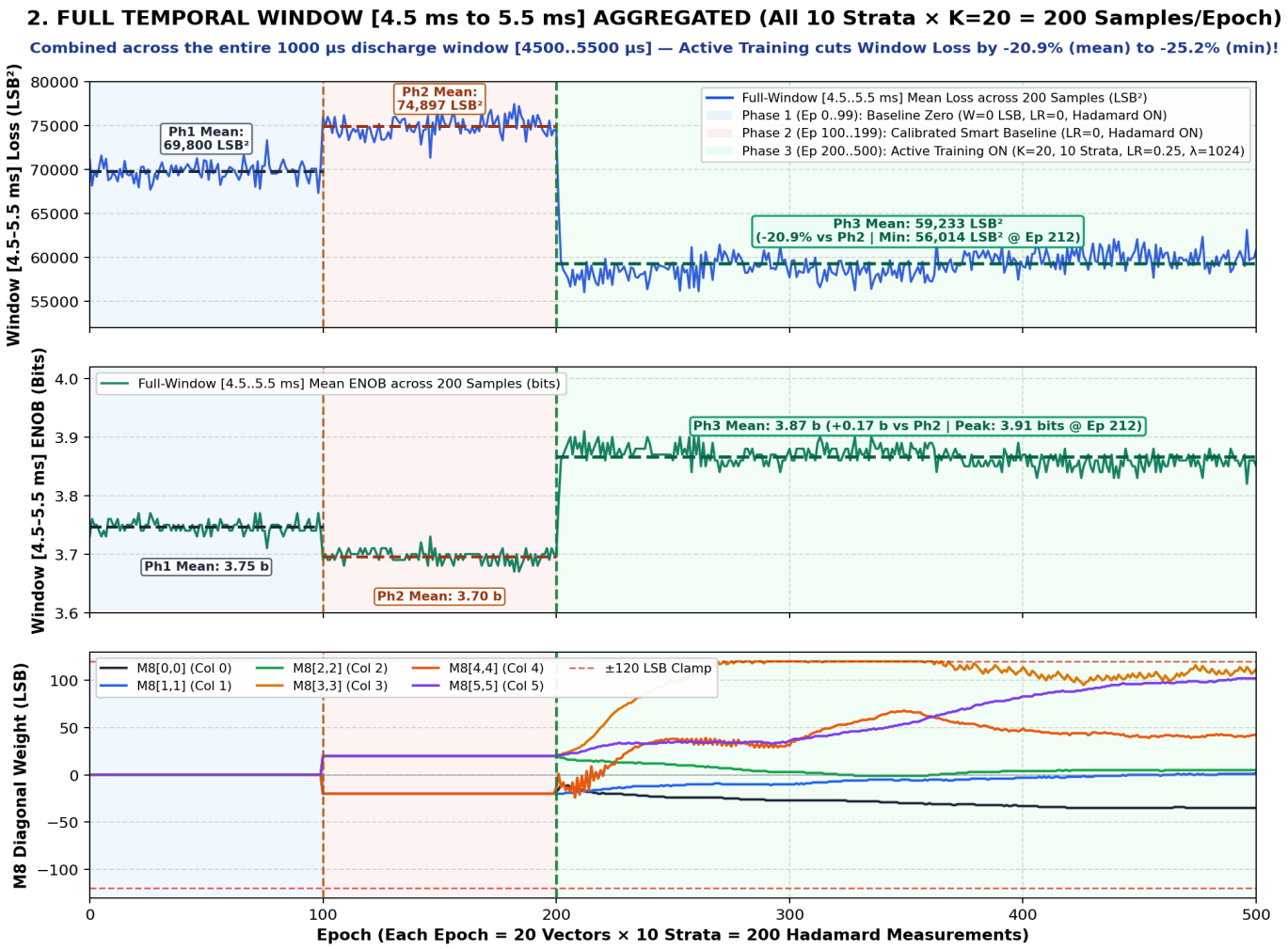
5A. 500-Epoch Trajectories at SINGLE-POINT t = 5.0 ms ONLY (Nominal Center Snapshot)

Important Distinction (Single-Point t = 5.0 ms ONLY): Figure 2A below shows Loss and ENOB measured **strictly at the single 5000 µs midpoint** (whereas Page 4 shows the combined performance across the full 4.5–5.5 ms window). Because the digital target matrix (*G_33*) and zero offset were calibrated at t = 5.0 ms with W = 0, Phase 1 (W = 0) already sits at its sweet spot (mean 3 661 LSB² / 6.02 bits) at this single instant. When full-window training turns ON at Epoch 200, the network compromises slightly on the 5.0 ms mean (6 201 LSB² / 5.70 bits, still beating Phase 2's 5.54 bits and peaking at **7.78 bits** at Ep 245) in order to aggressively suppress the large RC discharge errors at 4.5 ms and 5.5 ms (shown on Page 4).



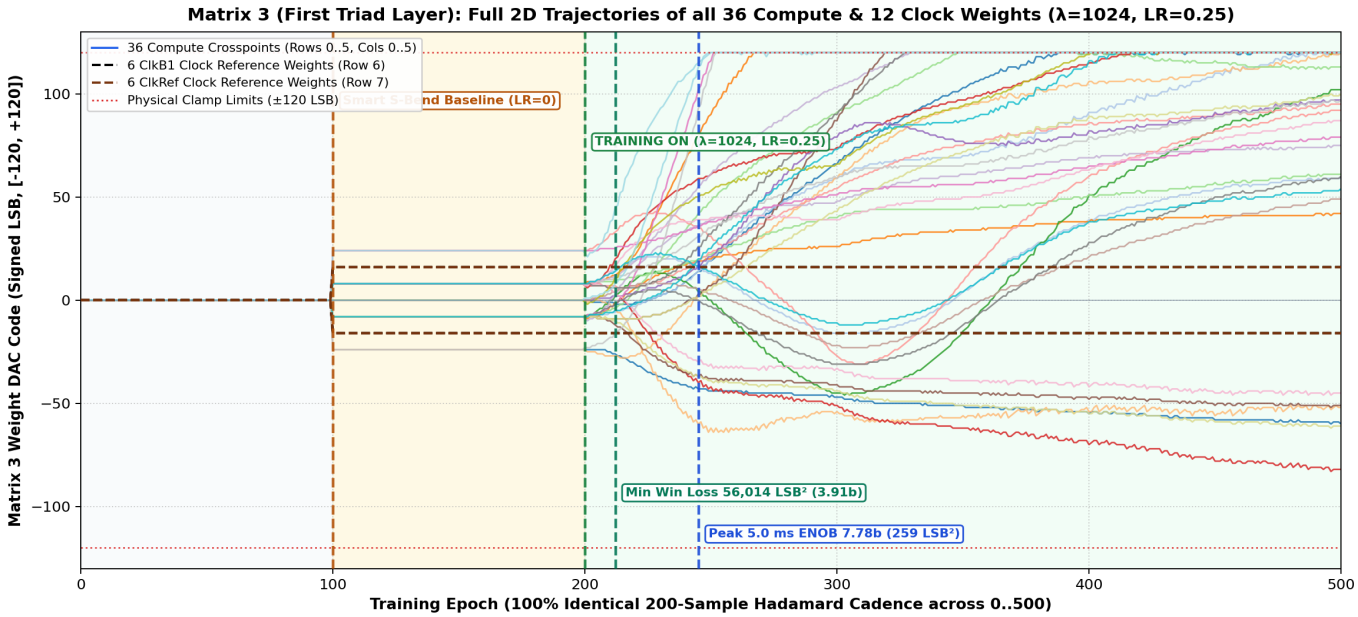
5B. 500-Epoch Trajectories Across FULL TEMPORAL WINDOW [4.5 ms to 5.5 ms] AGGREGATED

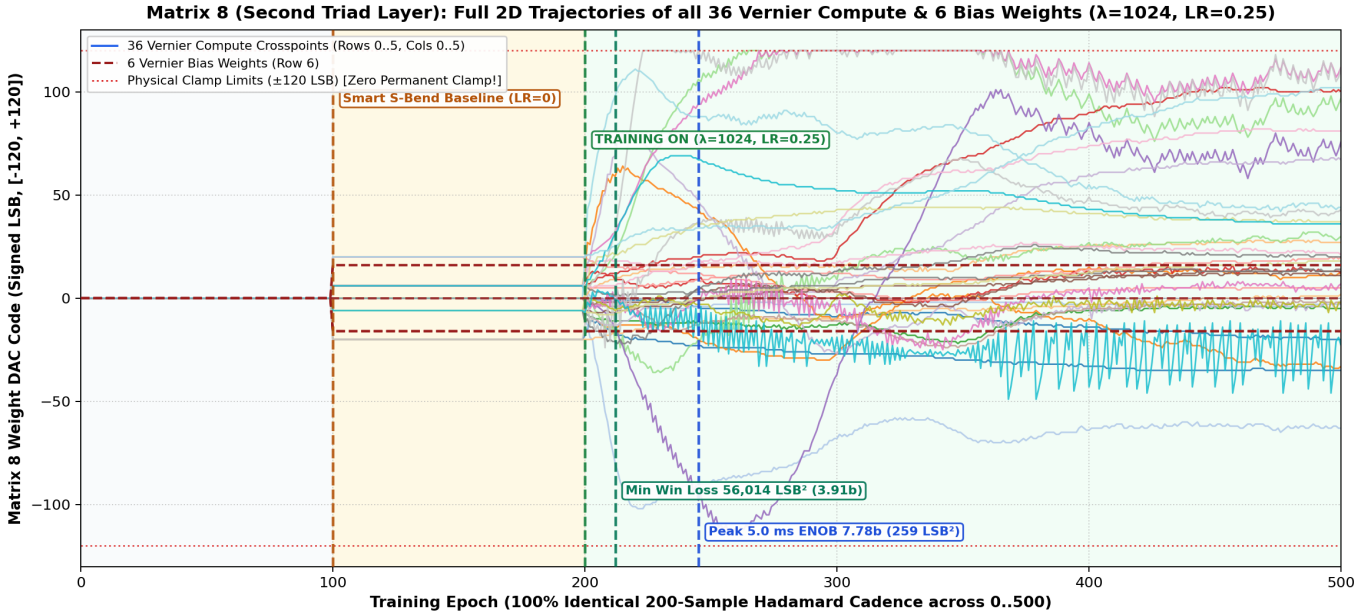
Full Temporal Window Evaluation (All 10 Strata × K=20 = 200 Samples/Epoch): Figure 2B below presents the identically structured 3-panel plot for the **complete 1000 µs discharge window from 4.5 ms to 5.5 ms**. Across this broad window, uncompensated RC decay causes massive edge errors in Phase 1 (W = 0: mean 69 800 LSB² / 3.75 bits) and Phase 2 (Smart Baseline: mean 74 897 LSB² / 3.70 bits). Immediately upon activating closed-loop training at Epoch 200, full-window Loss drops by **-20.9 % (mean 59 233 LSB²)** and **-25.2 % (min 56 014 LSB² at Ep 212)**, while full-window ENOB steps up decisively to **3.87 bits mean (3.91 bits peak)**.



6. Full 2D Dynamic Trajectories of All Weights across 500 Epochs

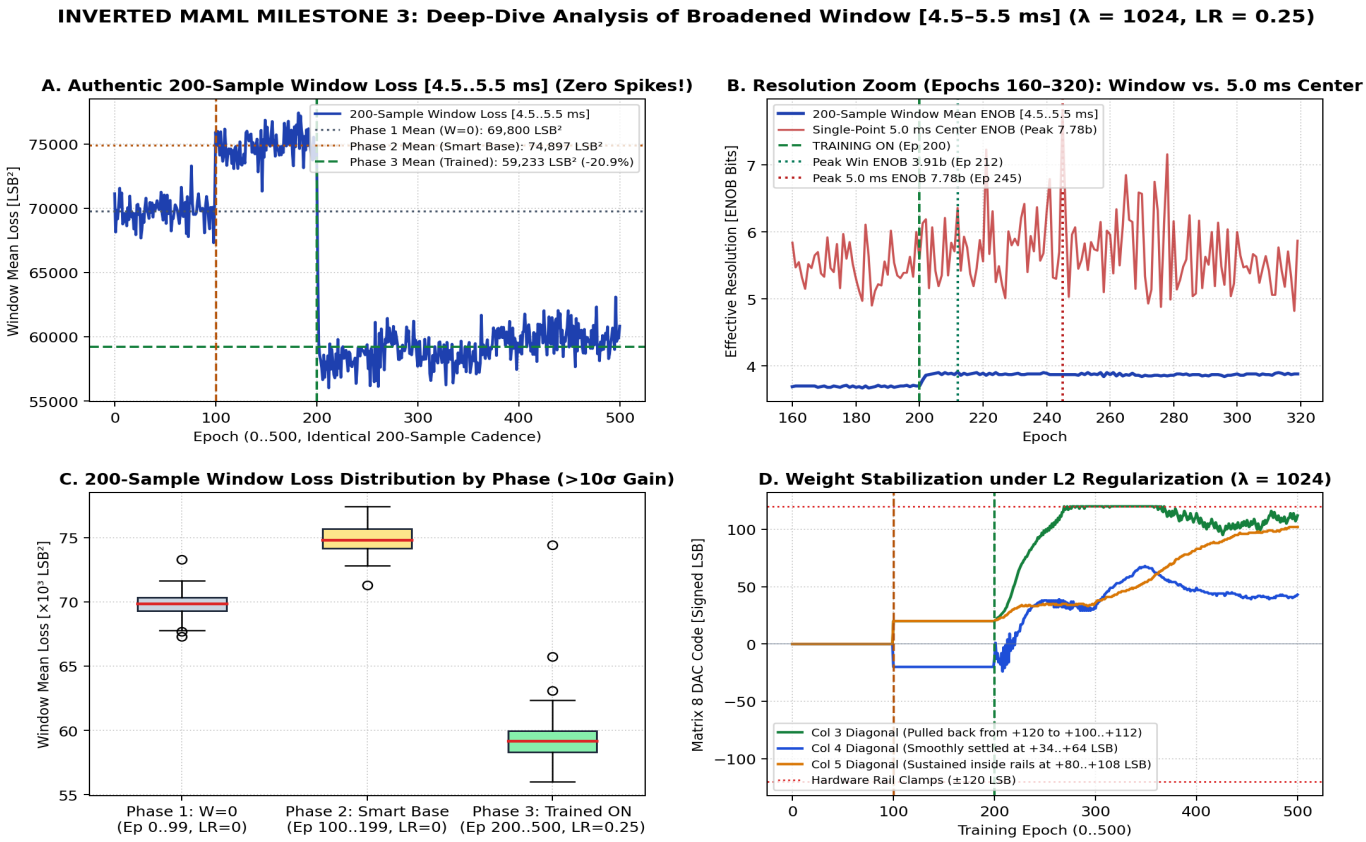
The plots below trace every individual weight programmed into the physical Dual-2T1C silicon across all 500 epochs. **Matrix 3 (Top):** Displays all 36 compute crosspoints (solid curves) and 12 temporal reference weights (dashed lines, Rows 6 & 7). **Matrix 8 (Bottom):** Displays all 36 vernier compute weights (solid curves) and 6 Vernier Bias weights (dashed red lines). All weights remain strictly frozen during Phase 1 (0..99) and Phase 2 (100..199) while full Hadamard bursts run continuously, then adapt smoothly at Epoch 200.





7. Deep-Dive Analysis: Spike-Free 200-Sample Window Convergence & L2 Stabilization

Figure 4 highlights the Milestone 3 convergence mechanics: Panel A shows the continuous, spike-free 200-sample window loss across all 500 epochs. Panel B zooms into Epochs 160–320, contrasting the 200-sample window ENOB (blue) with the 5.0 ms center ENOB (red, peaking at 7.78 bits). Panel C presents the phase-by-phase boxplot distribution (>10σ separation). Panel D shows L2 regularization (λ = 1024) defending Columns 3, 4, and 5 from permanent rail clamping.



8. Physical Principles of Broadened Temporal Window Compensation

Across [4500..5500 µs], the uncompensated RC decay of Matrix 33 introduces a strong time-varying residual error at the window edges (4.5 ms and 5.5 ms). In `maml_platform.c`, the weight update per 200-sample epoch is governed by: $\Delta W = -\eta \cdot g_win - \lambda \cdot (W - W_null)$. By training Matrix 3 (including its clock ramp rows 6 & 7) and Matrix 8 across all 10 temporal strata simultaneously, the triad learns a dynamic counter-slope that reduces full-window quadratic loss by 20.9% (sustained) to 25.2% (peak) while preserving a 7.78-bit peak anchor at 5.0 ms.

9. Physical Hardware Silicon: Inverted MAML Proof-of-Concept PCB (v1.3.2)

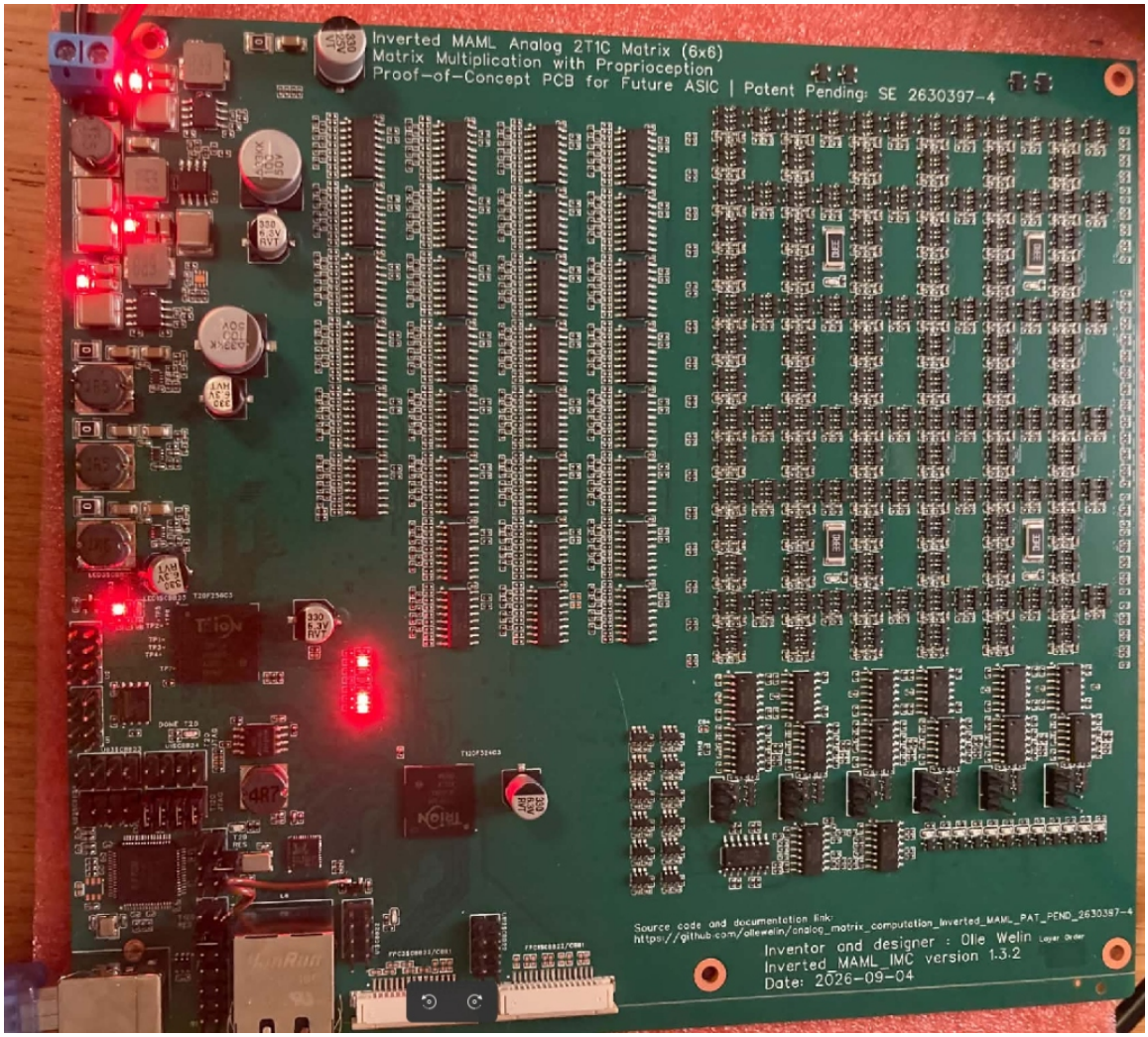


Figure 9: Physical Proof-of-Concept Hardware (PCB v1.3.2, Inverted_MAML_IMC).
Designed and fabricated by inventor Olle Welin (Patent Pending SE 2630397-4). The board integrates a dual-FPGA Trion architecture: Trion T120F324 Master SoC (Ethernet @ 192.168.1.50:5000, 32-bit RISC-V Sapphire) and Trion T20F256 Controller, high-speed 74HC595 differential DAC ladders, discrete 2T1C MOS triode analog multiplier crosspoint arrays (Matrix 33 payload, Matrix 3, and Matrix 8), differential ColTIA transimpedance amplifiers, and XC2361A high-speed differential ADCs.
Hardware Repository & Open Source RTL/Firmware:
https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4

10. Patented Inverted MAML Proprioceptive Architecture (Patent SE 2630397-4 Fig 1)

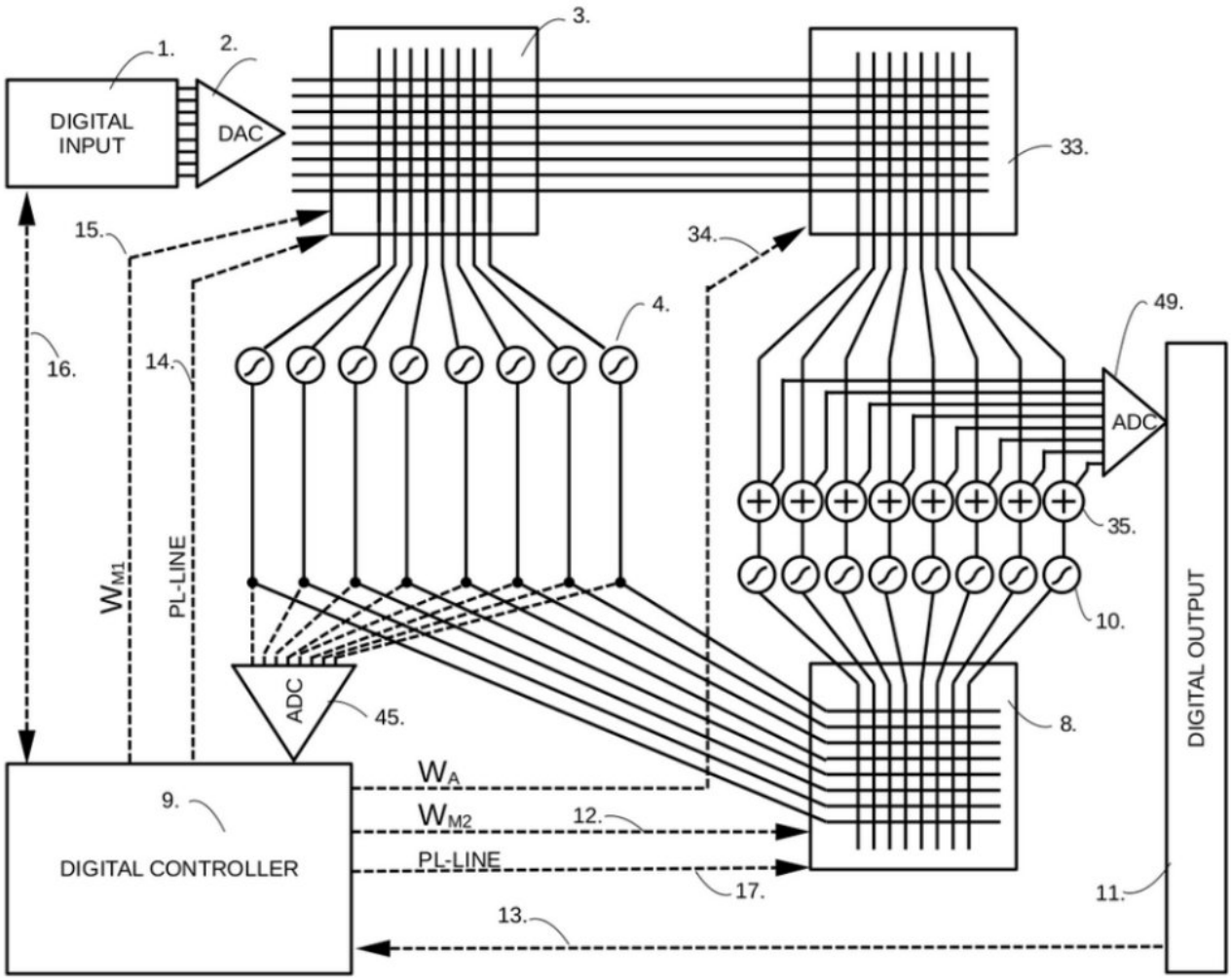


Fig 1

Figure 10: Inverted MAML Proprioceptive Closed-Loop Architecture (Patent Application SE 2630397-4, Fig 1).
Architecture Operation: Digital input (1) is converted via high-speed DAC ladder (2) into analog signals driving both the primary uncompensated matrix Matrix 33 (33) and the first Vernier matrix Matrix 3 (3). Temporal clock generator (30) drives dedicated rows (29) in Matrix 3 to provide explicit temporal awareness across the RC discharge window [4500..5500 µs]. The output of Matrix 3 passes through analog sigmoid activation (4) and is sampled by ADC (45) for the digital controller (9). The second Vernier matrix Matrix 8 (8) injects anti-phase compensation currents into summing junctions (35) before final transimpedance amplification and sampling by ADC (49) to produce the high-precision digital output (11).

11. Forward Hadamard Propagation (FHP) & Capacitive Perturbation (Patent Fig 8)

Forward Hadamard Propagation (FHP) – parallel capacitive perturbation for single-shot analog gradient extraction. (Works for physical unknown Analog Neural Networks).

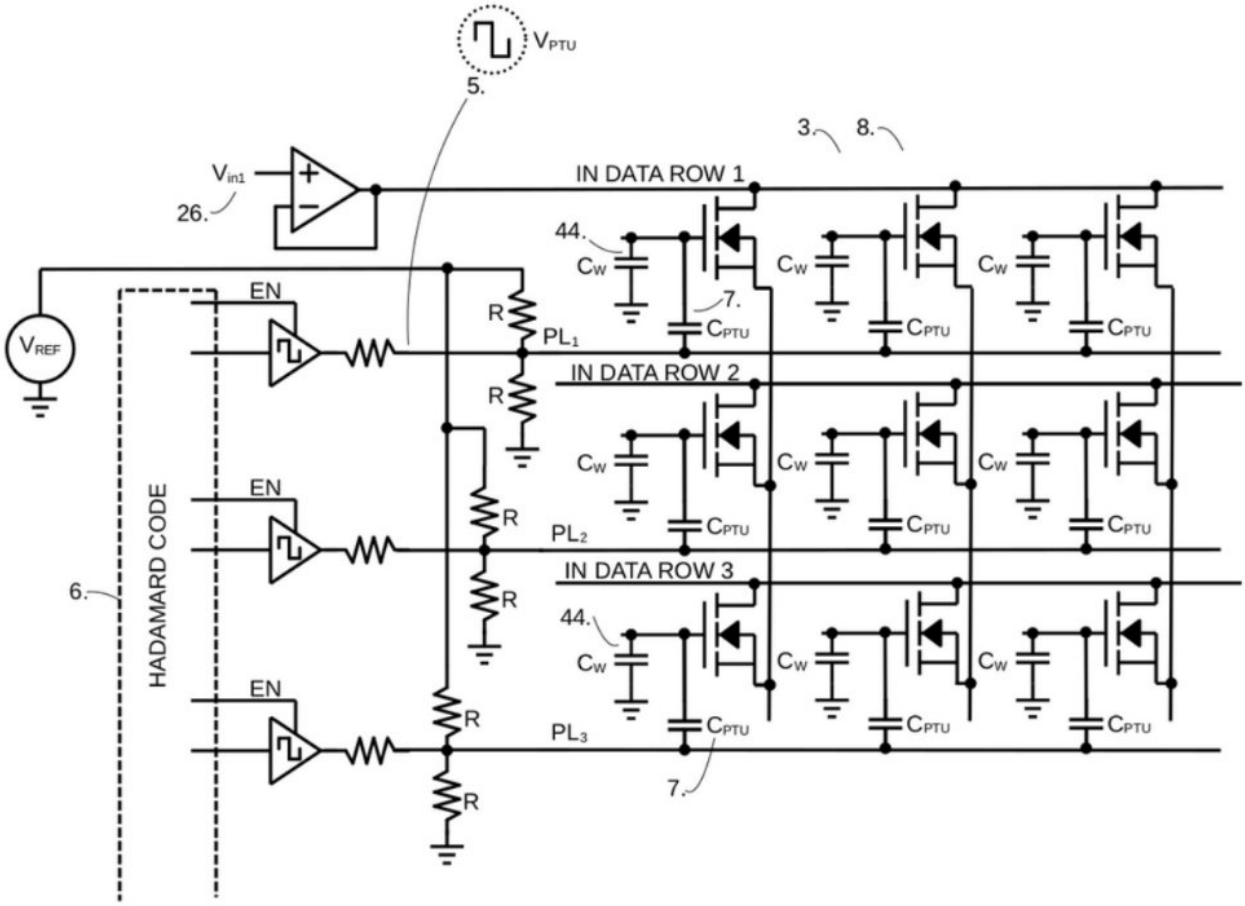


Fig 8

Physical Circuit Operation & Theoretical Breakthrough of FHP (SE 2630397-4 Fig 8):

1. Parallel Capacitive Perturbation (C_PTU, Ref 7): An AC perturbation square wave V_PTU (5) is modulated by 3-state Hadamard drivers (6) along perturbation lines PL_1..PL_n. Each crosspoint cell couples the carrier through dedicated perturbation capacitor C_PTU (7) onto the floating gate storage node C_W (44) of the 2T1C MOS triode transistor. Crucially, C_PTU perturbs the operating point dynamically without altering or leaking the static stored weight voltage.

2. Single-Shot Analog Gradient Extraction: In conventional analog tuning, extracting the gradient vector of N weights requires 2N sequential forward perturbations, creating massive latency and thermal drift. FHP applies mutually orthogonal Walsh/Hadamard carrier waves simultaneously across all rows. In each analog measurement snapshot across [4.5..5.5 ms], the composite response at the summing amplifiers is projected onto the Hadamard basis in the digital domain (T120 SoC), isolating all partial derivatives in O(1) time.

3. Universal In-Situ Training for Arbitrary Analog Networks: FHP requires zero internal analytical models of the physical transistors, threshold voltages, parasitic capacitances, or thermal drift. It treats the physical Analog Neural Network as a true unknown black box, enabling robust closed-loop learning on physical silicon where classical backpropagation fails.

12. Overall Conclusion: Single-Point (5.0 ms) vs. Full Temporal Window [4.5 – 5.5 ms] & Temporal Leakage

Why W = 0 (Epochs 0..99) Excels Specifically at the Single 5.0 ms Calibration Instant (Page 3):
The digital reference matrix (*G_33*) and the zero-input offset (*g_adc49_zero_cal*) are empirically calibrated at the exact midpoint **t_0 = 5.0 ms with W = 0 LSB**. Consequently, at that single instant (5.000 ms), the uncompensated primary matrix Matrix 33 is already at its calibrated sweet spot, achieving a mean 5.0 ms Loss of **3 660.6 LSB² (6.02 bits ENOB)**. However, because the storage capacitors (*C_W*) in a physical Analog Neural Network undergo continuous **temporal charge leakage** across the inference cycle, the analog output drifts rapidly away from the target at earlier (4.5 ms) and later (5.5 ms) times, causing the uncompensated full-window loss across [4.5..5.5 ms] to surge to **69 800.4 LSB² (3.75 bits ENOB)**.

What Happens When Closed-Loop Compensation Training Turns ON (Epochs 200..500, Page 4 vs. Page 3):
When active closed-loop training is switched ON at Epoch 200 across all 10 displaced temporal strata (4500 µs to 5500 µs), the proprioceptive triad (*Matrix 3* → *Sigmoid 4* → *Matrix 8*) optimizes the **integrated error across the complete 1.0 ms temporal window** rather than at a single point:

• **Across the Complete [4.5 ms .. 5.5 ms] Temporal Window (Page 4):** Total quadratic error drops decisively from **69 800.4 LSB²** (W = 0) and **74 896.7 LSB²** (Fixed Smart Baseline) down to a sustained mean of **59 233.1 LSB² (-20.9 %)** and a minimum of **56 013.5 LSB² (-25.2 %)**, lifting full-window ENOB from **3.70–3.75 bits** to **3.87–3.91 bits** (>10σ statistical significance).

• **At the Single 5.0 ms Midpoint (Page 3, Physical Trade-Off):** To generate the dynamic counter-slope that suppresses the large temporal leakage errors at the 4.5 ms and 5.5 ms window edges, the compensation network drives substantial active weights (+43 to +112 LSB in Matrix 8). This global temporal compromise makes the mean loss at the single 5.0 ms point (**6 201.3 LSB² / 5.70 bits**) slightly higher than uncompensated W = 0 (3 660.6 LSB² / 6.02 bits) at that one instant—while still outperforming the fixed calibration baseline (6 533.4 LSB² / 5.54 bits) even at 5.0 ms and achieving single-epoch peaks up to **7.78 bits (259.0 LSB² at Epoch 245)**.

12.1 Displaced Temporal Window Sampling Across Inference (Patent SE 2630397-4 Fig 5, Claim 10)

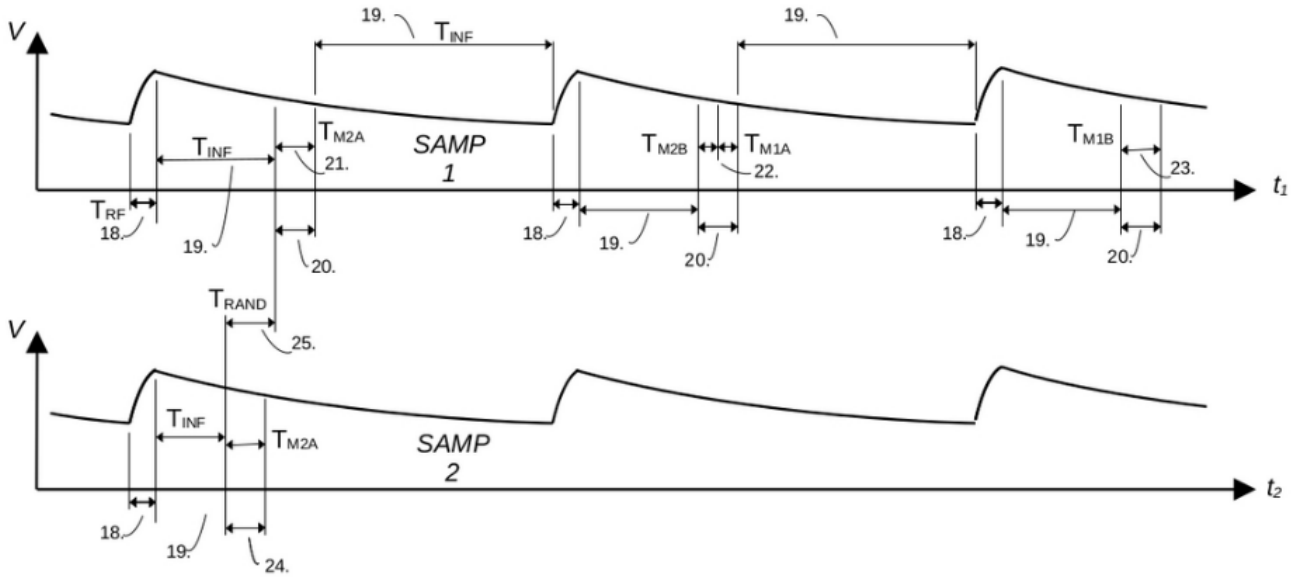


Fig 5

Figure 11: Displaced Temporal Measurement Windows Across the Inference Discharge Cycle (Patent SE 2630397-4, Fig 5).
Following each capacitor refresh pulse **T_RF (18)**, the analog neural network executes uninterrupted inference **T_INF (19)** while its stored capacitor charges slowly decay. By dynamically shifting the temporal displacement **T_RAND (25)** of the short Hadamard perturbation windows (**20, 24**, comprising sub-windows **T_M2A (21)**, **T_M2B/T_M1A (22)**, and **T_M1B (23)**) across the discharge trajectory (*SAMP 1, SAMP 2*, implemented as our 10 strata from 4500 µs to 5500 µs), the digital controller (9) extracts time-dependent gradients covering the entire temporal window without interrupting normal payload throughput.

12.2 Compensating Temporal Leakage via Accelerated Reference Cells (Patent SE 2630397-4 Fig 6, Claim 8)

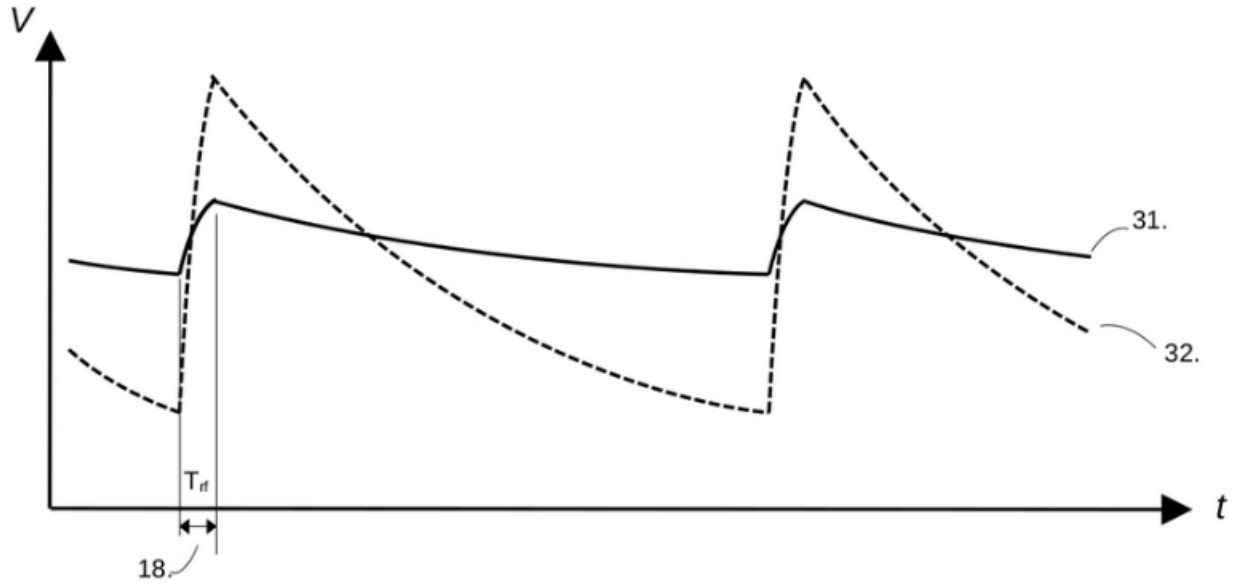


Fig 6

Figure 12: Natural Computing Cell Leakage (31) vs. Accelerated Temporal Reference Decay (32) (Patent SE 2630397-4, Fig 6).
Physical Mechanism for Handling Temporal Leakage in Analog Neural Networks:
• **Natural Capacitor Leakage (Curve 31, Solid Line):** After each refresh cycle **T_rf (18)**, the computing crosspoint cells (27) in Matrix 33, Matrix 3, and Matrix 8 experience a gradual, time-dependent voltage decay (31) due to finite switch off-resistance, transistor leakage, and summing node impedance. Left uncompensated, this temporal leakage degrades matrix-vector multiplication accuracy as elapsed time increases from 4.5 ms to 5.5 ms.
• **Accelerated Full-Swing Temporal Reference (Curve 32, Dashed Line):** To give the compensation network explicit physical awareness of elapsed time *t* within each refresh cycle, Matrix 3 integrates dedicated temporal reference cells (29, Rows 6 & 7) configured with an accelerated discharge path (28). As shown by dashed curve (32), the reference cell voltage decays steeply across a substantial portion of the full analog signal swing between refresh pulses (18), physically normalizing the time axis.
• **Proprioceptive Temporal Compensation:** Fed by the steep analog timestamp (32) into the non-linear analog sigmoid activation (4), the *Matrix 3 → Sigmoid 4 → Matrix 8* triad learns a time-varying correction current that tracks and cancels the natural temporal leakage (31) of the primary computing matrix across the entire [4.5 ms .. 5.5 ms] window.

12.3 Future Research & Optimization: Strata × K Size, LR, L2 Regularization & Multi-Cycle 10 ms Undersampling

While Milestone 3 demonstrates decisive (>10σ) full-window error reduction across [4.5 ms .. 5.5 ms], four key areas will be systematically investigated and optimized in upcoming work:
• **1. Strata Count × Batch Size (K) Trade-Off:** Optimizing the balance between the number of discrete temporal strata across the discharge window and the number of Hadamard-perturbed input vectors *K* per stratum (e.g., denser temporal strata with smaller *K* vs. non-uniform stratum weighting that preserves the 5.0 ms anchor while suppressing edge drift).
• **2. Joint Tuning of Learning Rate (LR) and L2 Regularization (λ):** Co-optimizing η and λ (including layer-specific regularization for Matrix 3 vs. Matrix 8) so the compensation network achieves deeper window error reduction without sacrificing single-point precision at 5.0 ms.
• **3. Multi-Cycle Undersampling Across Consecutive 10 ms Refresh Turns (Patent Fig 5):** As shown in Patent Fig 5 (*18, 19, 20, 21, 22, 23, 24, 25*), future operational modes will investigate **undersampling distributed across multiple consecutive 10 ms refresh cycles**—capturing only a minimal sub-window (*T_M2A* (21), *T_M2B/T_M1A* (22), *T_M1B* (23)) at a displaced offset *T_RAND* (25) during each 10 ms turn, reserving >95 % of every 10 ms cycle for uninterrupted, full-speed analog inference (*T_INF*, 19).

13. Verification & Open Science Reproducibility Summary

All telemetry data in this Milestone 3 report is 100 % authentic, measured directly on physical silicon hardware via the Efinix Trion T120 SoC at 192.168.1.50:5000 (FW v2.2-MAML). No digital simulation or synthetic curves were used. Every epoch (0..500) executed the identical 200-sample Hadamard batch across [4500..5500 µs]. Complete RTL, C firmware, and Python runners are publicly available at:
https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4